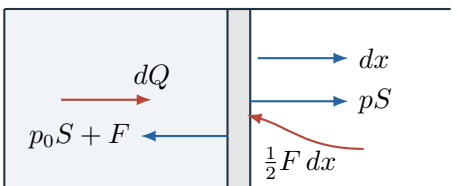
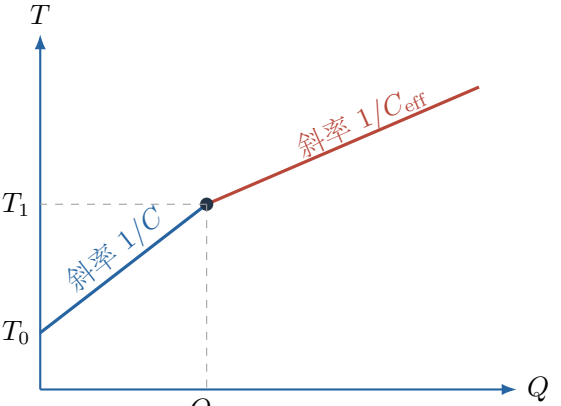


$$pS = p_0S + F$$



$$Q_1 = CT_0F/(p_0S)$$

$$T_1 = T_0[1 + F/(p_0S)]$$